

# Real-Time Obstacle Detection with Alert and Emergency Braking Using Arduino

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**Abstract**—This project presents the design of a distance measuring device utilising an Arduino Uno microcontroller and Ultrasonic sensor (HC-SR04). This equipment operates by emitting an ultrasonic wave and calculating the time to return it from the reflecting surface, allowing us to measure the distance accurately. This data is displayed on a 16×2 LCD in real time, providing a user-friendly experience. This project further indicates to the user whether the edge is closer to an obstacle, enabling an LED and a buzzer. This device could be used in cars, where it could act as a guide for reverse parking by providing them with real-time distance between the vehicle and the obstacle, and timely alerts ensure the vehicle's safety. For further protection, we added a feature where the motor would turn off automatically when the distance between the sensor and the wall is significantly less.

## I. INTRODUCTION

We all know that a lack of guidance while reverse parking is the reason for bumping into some obstacle or another car. This device gives the driver information about how far the obstacle is, and guides the driver with an LED and a buzzer. When the distance between the obstacle and the sensor is over 50 cm, the LED remains green and the buzzer stays silent. As we cross the mark of 50cm, the LED remains green but the buzzer starts to ring with less frequency, indicating to the driver that the car is now near the obstacle and should slow down. When the distance between the sensor and the obstacle is less than 20 cm, the device is programmed such that the LED turns red, the buzzers start to ring continuously, and the motor stops rotating to prevent any possible accident.

## II. MEASUREMENT OF DISTANCE USING ULTRASONIC SENSOR

The accuracy of the whole device depends on how accurately the device captures the distance of the obstacle from the sensor. We used an HC-SR04 ultrasonic sensor (Fig. 1) to calculate the distance. The ultrasonic sensor consists of two modules. One is a transmitter and the other is a receiver module. The transmitter module sends a pulse of ultrasonic waves at regular intervals, allowing us to track the distance in real time. The pulse sent reflects on the obstacle's surface, and the receiver receives the reflected pulse. The ultrasonic sensor calculates the time between the sent wave and when the receiver receives the reflected wave. The data we know now is the time taken for reflection and the speed of the ultrasonic wave in air. Using the formula  $\text{velocity} = \text{distance}/\text{time}$ , we can accurately calculate the distance between the sensor and the obstacle.



Fig. 1: Ultrasonic sensor HC-SR04

## III. DESCRIPTION OF THE PROTOTYPE

Fig. 2 shows the prototype of a device designed to assist drivers during reverse parking, intended to be mounted on the rear of the user's car. An ultrasonic sensor (HC-SR04) is being used to calculate the distance of the obstacle from the sensor. This data is then fed into the D6 and D7 pins of the Arduino board. Furthermore, this data is displayed on the LCD screen, which is connected to A4 and A5 pins of the Arduino board. Besides that, the RGB LED is connected to D10, D11, D12 pins, a buzzer to D8 pin, and a motor driver connected to D9 pin (which controls the motor's action on the microcontroller's commands).

When the obstacle is far from the sensor, the microcontroller allows it to continue without any issue. As the obstacle comes within 50 cm of the sensor, the microcontroller commands the buzzer to ring lowly and enables the user to proceed carefully. Suppose the obstacle is within 20 cm of the sensor. In that case, the microcontroller commands the LED to glow red, the buzzer to buzz continuously, and the motor driver to halt the motor immediately to avoid bumping into the obstacle.

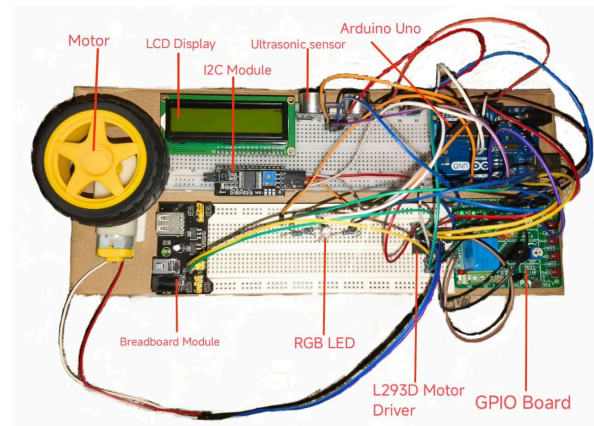


Fig. 2: Experimental Setup

#### IV. EXPERIMENTAL RESULT

##### A. Observation Table:

| Distance from obstacle<br>(in cm) | Buzzer<br>Status | LED<br>colour | Motor<br>status |
|-----------------------------------|------------------|---------------|-----------------|
| > 50                              | Off              | Green         | On              |
| >20 and <50                       | Slow Beep        | Green         | On              |
| <20                               | Continuos Beep   | Red           | Off             |

**Table 1: Results**

##### B. Accuracy:

The sensor measures the distance of the obstacle from the sensor with marginal errors. Hence, this prototype is expected to work pretty accurately without damaging the car in a real scenario.

##### C. Limitations:

Very fast fluctuations in readings due to uneven surface and instability of the hand while experimenting. This sensor is designed to work efficiently in a range of 2cm to 400cm, which limits the usage of this prototype.

#### V. CONCLUSION

Arduino Uno-based distance measurer and alert system with braking mechanism provides a practical and effective solution for enhancing safety. We conclude that this project demonstrated how an ultrasonic sensor can be utilised for real-time distance monitoring and how microcontrollers can prevent collisions and ensure a timely response to obstacles. This project highlights its potential for further development in the automation field, paving the way for safer integration with cars.

#### VI. FUTURE SCOPE

- I. Replacing the Ultrasonic sensor with a LiDAR sensor to get accurate results and better range
- II. Adding voice command for better communication with the driver (like Google Maps).

#### ACKNOWLEDGEMENT

The authors acknowledge the Electrical Lab at the Indian Institute of Technology, Gandhinagar, for providing the necessary equipment and sensors to develop this prototype. We also sincerely thank Ms. Shipra Singh for her continuous guidance throughout this project.

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